

Rolle's and MVT

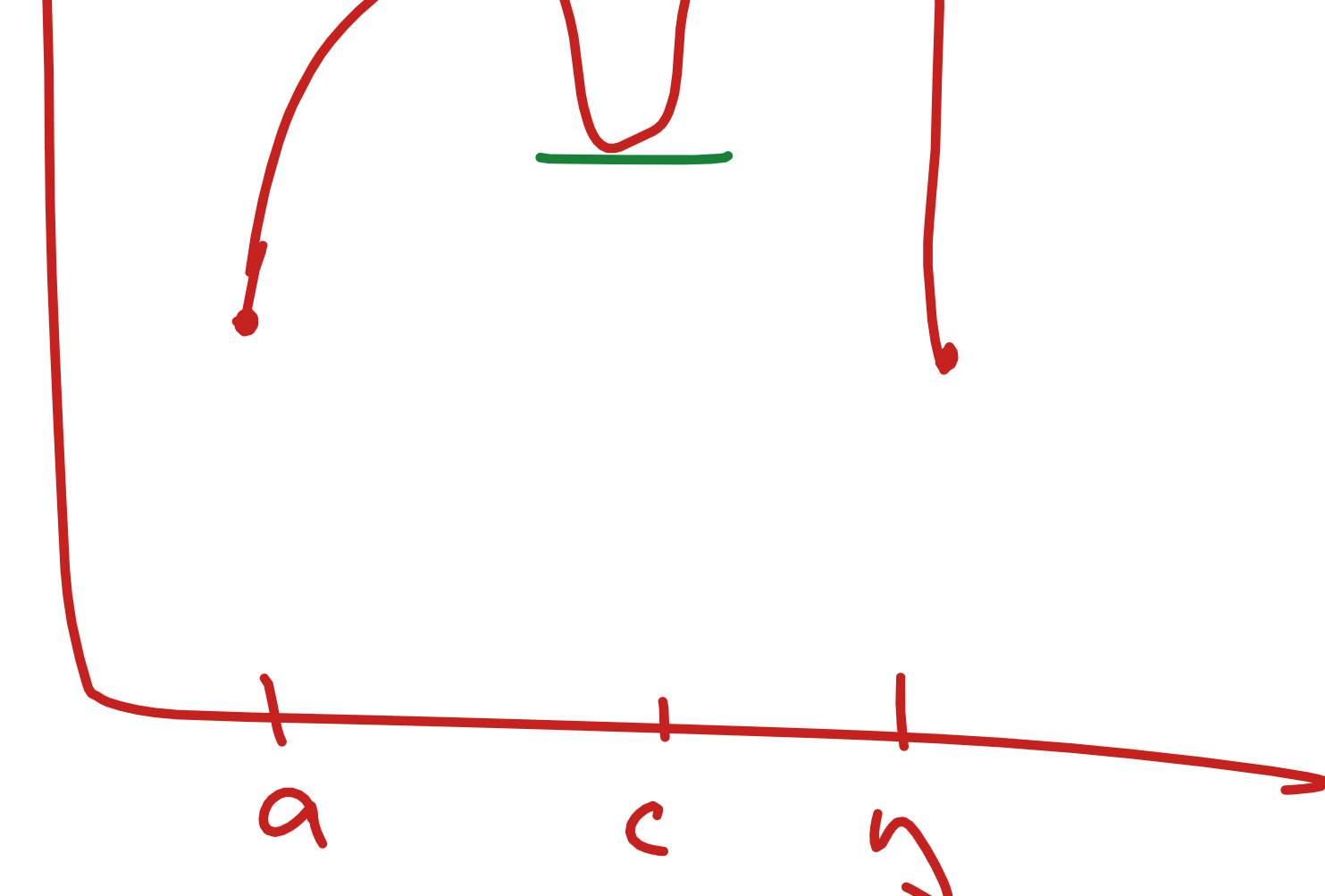
section 3.2

9/23/25

A: Rolle's

conditions

- 1) $f(x)$ is continuous on $[a, b]$
- 2) $f(x)$ is differentiable on (a, b)
- 3) $f(a) = f(b)$



Rolle's guarantees at least one time in the interval, $f'(x) = 0$

Ex:

$$f(x) = x^{1/3} \quad [-2, 2]$$

- 1) $f(x)$ is continuous on $[-2, 2]$
- 2) $f(x)$ is not differentiable on $(-2, 2)$

Rolle's does not guarantee a time when $f'(x) = 0$ on $(-2, 2)$

$$\text{Ex 2: } f(x) = 3x + 2$$

- 1) $f(x)$ is continuous on $[0, 1]$
- 2) $f(x)$ is differentiable on $(0, 1)$
- 3) $f(0) = 2, f(1) = 0, 0 \neq 2, f(0) \neq f(1)$ ✓

Rolle's doesn't guarantee a time when $f'(x) = 0$ on $(-2, 2)$

Ex 3:

$$f(x) = \sin x \quad [0, \pi]$$

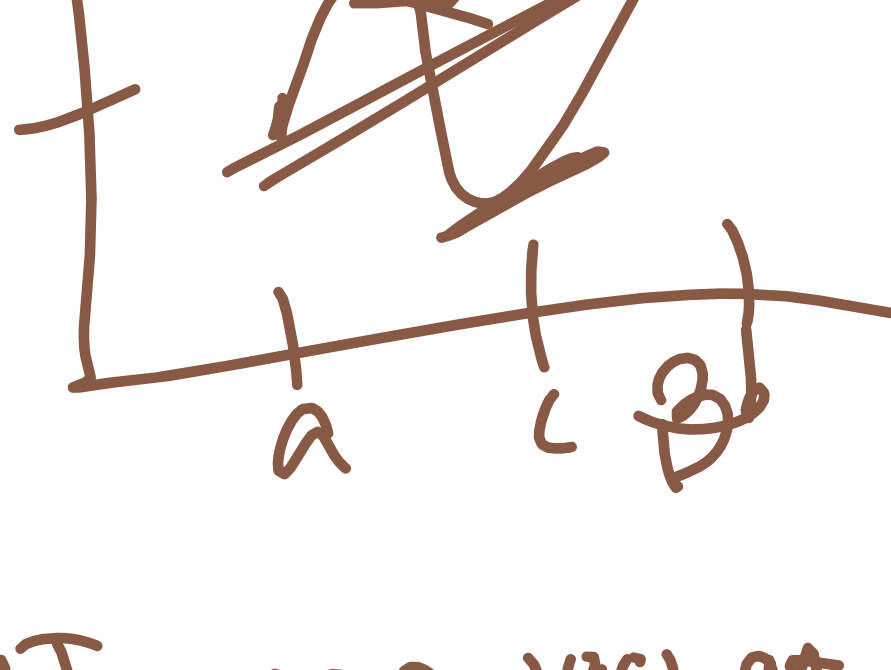
~~$$f'(x) = \cos x$$~~

- 1) $f(x)$ is continuous on $[0, \pi]$
- 2) $f(x)$ is differentiable on $(0, \pi)$
- 3) $f(0) = 0, \sin(\pi) = 0, f(0) = f(\pi)$

Rolle's guarantees that at least one time on $[0, \pi], f'(x) = 0$

Mean Value Theorem (MVT)

- 1) $f(x)$ is continuous on $[a, b]$
- 2) $f(x)$ is differentiable on (a, b)
- 3) $\frac{f(b) - f(a)}{b - a} = m$



MVT guarantees at least one time in the interval when $f'(x) = m$

$$f(x) = x^2 \quad [0, 2]$$

- 1) $f(x)$ is continuous on $[0, 2]$
- 2) $f(x)$ is differentiable on $(0, 2)$

$$3) \frac{f(2) - f(0)}{2 - 0} = 2$$

MVT guarantees at least one time

in $(0, 2)$ when $f'(x) = 2$

$$\boxed{f'(x) = 2}$$

Solve for x

$$2x = 2$$

$$\boxed{x = 1}$$